

energy $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-2}^2 25 dt = 100;$

average power $P_x = \lim_{T \rightarrow \infty} \frac{1}{T} E_x = 0.$

Energy Signal

average power $P_z = \frac{1}{8} \int_{-4}^4 |z(t)|^2 dt = \frac{1}{8} \int_{-2}^2 25 dt = \frac{100}{8} = 12.5;$

energy $E_z = \int_{-\infty}^{\infty} |z(t)|^2 dt = \infty.$

Power Signal

- A signal **can't** be both energy and power simultaneously.
- A signal may be neither energy nor power signal
- Generally **periodic** signals are **power** signals
- Generally **aperiodic** signals are **energy** signals

In Conclusion :

۱- اگر E_{∞} محدود باشد ، $P_{avg \infty}$ آن صفر است

۲- در سینیال متناوب توان کس سینیال ، بتوان کس در سینیال برابر است .

C.T. $P_{avg \infty} = \frac{1}{T_0} \int_{T_0} |x(t)|^2 dt$

D.T. $P_{avg \infty} = \frac{1}{N} \sum_{n \in \langle N \rangle} |x[n]|^2$

۳- انرژی کس سینیال غیر متناوب ، نامحدود است (∞)

Given $x(t)$ Find $x(\alpha t + \beta)$

(I) سینیال $x(t)$ به اندازه β انتقال داده شود

الف) $\alpha > 0$ انتقال به سمت راست

ب) $\alpha < 0$ انتقال به سمت چپ

۲ ضرب به سینیال داده می شود (تسک کردن)

الف) اعداد بر روی محور x بر α تسک می شود

ب) در صورت منفی بودن α سینیال نسبت به محور t تسک می شود

(II) درصد ضرب از به فاکتور تسک می شود

۱ ابتدا ضرب به سینیال داده می شود

الف) اعداد بر روی محور x بر α تسک می شود

۲ اگر α منفی بود سینیال نسبت به t تسک می شود

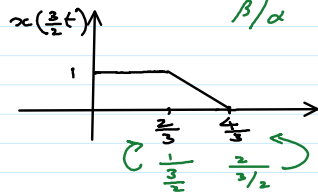
۳ ضرب به سینیال داده می شود

الف) $\alpha > 0$ به سینیال تسک می شود

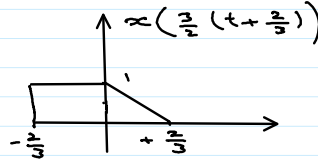
(I) $\beta < 0$ انتال به سمت راست

ϵ_n

$$x\left(\frac{3}{2}\left(t + \frac{2}{3}\right)\right)$$



$\frac{\beta}{\alpha} > 0$ shift to the left



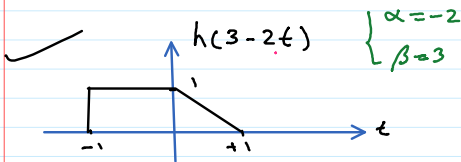
(II) نحوه بیت آوردن $x(t)$ در صورتی $x(\alpha t + \beta)$ را می بینیم

(1) ابتدا $|\alpha|$ را تعیین می کنیم

تمام اعداد در محور x را در $|\alpha|$ ضرب می شود

(2) در صورتی که α منفی بود

(3) عدد β را می بینیم و اگر $\beta > 0$ است (انتال به سمت راست) و اگر $\beta < 0$ است (انتال به سمت چپ)



Find $h(t)$

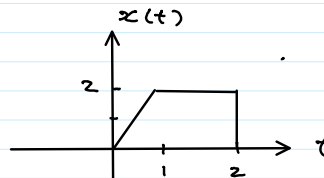
Exercice: $x(t)$ is given by

Find

$$A = \int_{-\infty}^{\infty} x\left(1 - \frac{t}{2}\right) dt$$

$$\alpha = 1 - \frac{t}{2}$$

$$= 5.5$$



$$d\alpha = -\frac{1}{2} dt$$

$$dt = -2 d\alpha$$

Periodic Signals :

$$g_1(t) = 3 \sin(4\pi t) + 7 \cos(3\pi t)$$

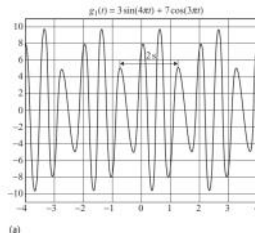
are both periodic signals with fundamental periods $1/2$ and $2/3$, respectively. Calculating the ratio of the two fundamental periods yields

$$\frac{T_1}{T_2} = \frac{1/2}{2/3} = \frac{3}{4}$$

3 & 4 are coprime (both are prime #)

$$4T_1 = 3T_2$$

$$T = 2$$

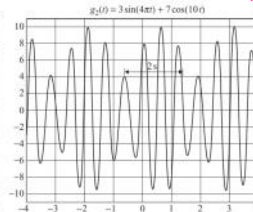


(a)

$$g_2(t) = 3 \sin(4\pi t) + 7 \cos(10\pi t)$$

signals with fundamental periods $1/2$ and $\pi/5$, respectively. Calculating the ratio of the two fundamental periods yields

$$\frac{T_1}{T_2} = \frac{1/2}{\pi/5} = \frac{5}{2\pi}$$



(b)

Fig. 1.8. Signals (a) $g_1(t)$ and (b) $g_2(t)$ considered in Example 1.5. Signal $g_1(t)$ is periodic with a fundamental period of 2 s, while $g_2(t)$ is not periodic.

Sum of periodic Signals

$$x(t) = x_1(t) + x_2(t) + \dots + x_M(t)$$

$$T_{01}$$

$$n_1 T_{01}$$

$$T_{02}$$

$$n_2 T_{02}$$

$$T_{0m}$$

$$n_m T_{0m}$$

T_{0i} = fundamental period of $x_i(t)$.
 $n_1 T_{01} = n_2 T_{02} = \dots = n_m T_{0m} = T_0$

$\frac{T_{01}}{T_{0i}} = \frac{n_i}{n_1} \quad 2 \leq i \leq m$
 $n-1$ ratios rational #

Procedure:

1. Convert each period ratio $\left(\frac{T_{01}}{T_{0i}}, 2 \leq i \leq m\right)$ to a ratio of integers,
2. Eliminate any common factors from numerator & denominator of each ratio (become coprime)
3. Find least common multiple (LCM) of the denominators of the ratios. (This value is n_1)
4. Compute the fundamental period of $x_s(t)$ as $T_0 = n_1 T_{01}$

$x_1(t)$ is periodic $T_{01} = 1.08$ Sec.

$x_2(t)$ " " $T_{02} = 3.6$ Sec

$x_3(t)$ " " $T_{03} = 2.025$ Sec

is $x(t) = x_1(t) + x_2(t) + x_3(t)$ periodic?

$$\frac{T_{01}}{T_{02}} = \frac{1.08}{3.6} = \frac{108}{360} = \frac{3}{10}$$

$$\frac{T_{01}}{T_{03}} = \frac{1.08}{2.025} = \frac{1080}{2025} = \frac{8}{15}$$

$LCM(10, 15) = 30$

$$T_0 = 30 T_{01} = 32.4$$

is $f[n] = \cos\left(\frac{8\pi n}{31}\right)$ periodic?

$$f[n] = f[n + N]$$

Find N

s.t. $\cos\left(\frac{8\pi(n+N)}{31}\right) = \cos\left(\frac{8\pi n}{31}\right)$

$$F_{in} \perp N$$

s.t.

$$\cos\left(\frac{8\pi(n+N)}{31}\right) = \cos\left(\frac{8\pi n}{31}\right)$$

using periodicity of the cosine function, which has a period of 2π , this leads to:

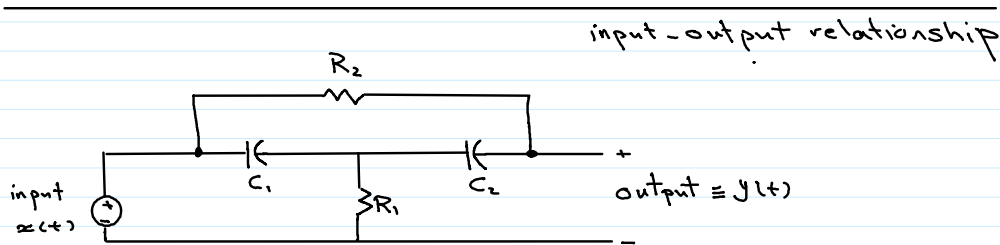
$$\frac{8\pi N}{31} = 2\pi k \text{ for some integers of } k.$$

simplifying $\frac{8N}{31} = 2k$ or $N = \frac{31}{4}k$

To find the smallest integer N, we need k to be a multiple of 4 for N to be an integer.

The smallest integer value for k is 4, which gives

$$\frac{(31)(4)}{4} \Rightarrow \boxed{N=31}$$



$$R_1 C_1 \frac{d^2 y(t)}{dt^2} + \left(1 + \frac{R_1 C_1}{R_2 C_2} + \frac{R_1}{R_2}\right) \frac{dy(t)}{dt} + \frac{1}{R_2 C_2} y(t) =$$

$$R_1 C_1 \frac{d^2 x(t)}{dt^2} + \left(\frac{R_1 C_1}{R_2 C_2} + \frac{R_1}{R_2}\right) \frac{dx(t)}{dt} + \frac{1}{R_2 C_2} x(t)$$

$$a_2 \frac{d^2 y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_2 \frac{d^2 x(t)}{dt^2} + b_1 \frac{dx(t)}{dt} + b_0 x(t)$$

input-output of a system is given

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]$$

stability: Since for bounded input like $x[n] = 1$

$$y[n] = \sum_{k=-\infty}^{+\infty} (1) = \infty \text{ is unstable}$$

Stability: Since for bounded input like $x[n] = 1$

$$y[n] = \sum_{k=n-1}^{+2} (1) = \infty \quad \text{is unstable}$$

Causality: Since output depends on future values of input is NOT Causal system

invertibility: First we find $y[n+2]$ & $y[n+1]$

$$\begin{aligned} y[n+\tau] &= \sum_{k=n+1}^{+\infty} x[k], \quad y[n+1] = \sum_{k=n}^{+\infty} x[k] \\ y[n+1] - y[n+\tau] &= \left(\sum_{k=n}^{+\infty} x[k] \right) - \left(\sum_{k=n+1}^{+\infty} x[k] \right) = x[n] \\ x[n] &= y[n+1] - y[n+\tau] \end{aligned}$$

Since $x[n]$ can be find from $y[n]$, it is invertible

Time-invariance:

For $x_{[n]}$ the output is:

$$y_1[n] = \sum_{k=n-1}^{+2} x_1[k] \quad \text{معادله (1)}$$

حالت ورودی $x_1[n]$ را به اندازه n جابجا می‌کنیم و ورودی جدید $x_1[n] = x_1[n-n_0]$ را به سیستم اعمال می‌کنیم. خروجی $y_1[n]$ برابر است با:

$$y_r[n] = \sum_{k=n-1}^{+\infty} x_r[k] = \sum_{k=n-1}^{+\infty} x_r[k-n_r] \xrightarrow{k'=k-n_r} y_r[n] = \sum_{k'=n-n_r-1}^{+\infty} x_r[k']$$

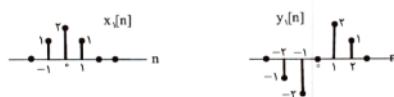
بنابراین معادله $y_T[n] = \sum_{k=n-n_-}^{+\infty} x_U[k]$ برقرار است. در ادامه در معادله ۱، $y_U[n-n_-]$ را تشکیل می‌دهیم.

$$(1) \text{ معادله } \Rightarrow y_1[n-n_1] = \sum_{k=n-n_1-1}^{+\infty} x_1[k] \quad (3) \text{ معادله}$$

سیستم تغییرناپذیر با زمان است $\Rightarrow y_1[n] = y_1[n - n_s] \Rightarrow$ معادله (۳) و (۴)

LTI Systems:

input $x, [n]$ output $y, [n]$ given below:



Find its response when $x_2[n]$ is



Solution:

Comparing $x_i[n]$ & $y_i[n]$, we see the following relationship

$$y_1[n] = x_1[n-1] - x_1[n+1]$$

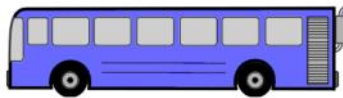
$\therefore$ we can conclude,

$$y[n] = x[n-1] - x[n+1]$$

use this to find the output

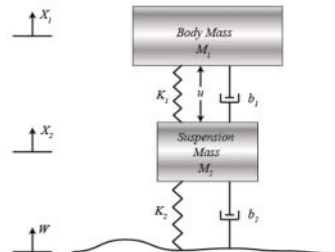
Automotive Suspension System

Physical setup



Designing an automotive suspension system is an interesting and challenging control problem. When the suspension system is designed, a 1/4 model (one of the four wheels) is used to simplify the problem to a 1-D multiple spring-damper system. A diagram of this system is shown below. This model is for an active suspension system where an actuator is included that is able to generate the control force U to control the motion of the bus body.

Model of Bus Suspension System (1/4 Bus)



System parameters

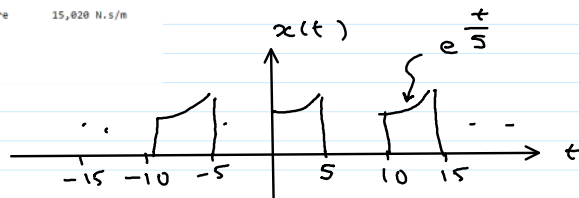
(M1)	1/4 bus body mass	2500 kg
(M2)	suspension mass	320 kg
(K1)	spring constant of suspension system	80,000 N/m
(K2)	spring constant of wheel and tire	500,000 N/m
(b1)	damping constant of suspension system	350 N.s/m
(b2)	damping constant of wheel and tire	15,020 N.s/m
(U)	control force	

Equations of motion

From the picture above and Newton's law, we can obtain the dynamic equations as the following:

$$M_1 \ddot{X}_1 = -b_1(\dot{X}_1 - \dot{X}_2) - K_1(X_1 - X_2) + U$$

$$M_2 \ddot{X}_2 = b_1(\dot{X}_1 - \dot{X}_2) + K_1(X_1 - X_2) + b_2(\dot{W} - \dot{X}_2) + K_2(W - X_2) - U$$



is periodic $T_0 = 10$ $\omega = \frac{2\pi}{T_0} = \frac{\pi}{5}$

$$x(t) = \sum_{k=-\infty}^{+\infty} C_k e^{j k (\frac{\pi}{5}) t}$$

ω

$$C_k = \frac{1}{T} \int_{\langle T \rangle} x(t) e^{-j k \omega t} dt = \frac{1}{10} \int_0^5 \left(e^{\frac{t}{5}} \right) e^{-j k \frac{\pi}{5} t} dt$$

$-P, +P$ $\langle T \rangle$

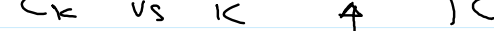
$$= \frac{1}{10} \frac{5}{1 - j k \pi} e^{\frac{t}{5} (1 - j k \pi) t} \Big|_0^5$$

$$C_k = \frac{e^{-(j k \pi - 1)} - 1}{2 - j 2 k \pi}$$

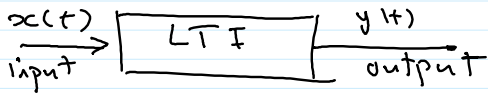
Complex $\neq$

Real part C_k vs k $\uparrow$ $|C_k|$ vs k
 τ n $+$ C $\dots$ x C_k vs k

Real part C_k vs k $|C_k|$ vs k
Im part C_k vs k $\angle C_k$ vs k



E_x



impulse Response $h(t) = e^{-4t} u(t)$

$$x(t) = \sin(4\pi t) + \cos(6\pi t + \frac{\pi}{4}) \quad \checkmark$$

I) Find C_{ik} for input?

$$\omega_{01} = \frac{2\pi}{T_{01}} \Rightarrow T_{01} = \frac{2\pi}{4\pi}$$

II) Find b_k for output?

Solution:

$$\begin{aligned}\cos(\alpha) &= \frac{1}{2} (e^{j\alpha} + e^{-j\alpha}) \\ \sin(\alpha) &= \frac{1}{2j} (e^{j\alpha} - e^{-j\alpha})\end{aligned}$$

also is periodic $T_0 = 1$ $\omega_0 = 2\pi$

$$\rightarrow x(t) = \sum_{k=-\infty}^{+\infty} C_k e^{j k (2\pi) t}$$

$$\rightarrow x(t) = \frac{1}{2j} \left(e^{j(2) \cdot 2\pi t} - e^{j(-2) \cdot 2\pi t} \right) + \frac{1}{2} \left(e^{j(3 \cdot 2\pi t + \frac{\pi}{4})} + e^{j((-3) \cdot 2\pi t - \frac{\pi}{4})} \right)$$

$$C_2 = \frac{1}{2}$$

$$C_{-2} = -\frac{1}{2}$$

$$C_3 = \frac{1}{2} e^{j\frac{\pi}{2}}$$

$$C_{-3} = \frac{1}{2} e^{-j \frac{\pi}{4}}$$

$$H. \quad y(t) = \sum_{k=-\infty}^{+\infty} C_k \cdot e^{j k \omega_0 t}$$

$$| -1(jk\omega_0) = H(\omega) |_{\omega = k\omega_0} = \int_{-\infty}^{+\infty} h(\tau) e^{-j\omega\tau} d\tau$$

$$= \int_0^{+\infty} \underbrace{e^{-4\tau} \cdot e^{-j\omega\tau}}_{e^{-(j\omega+4)\tau}} d\tau \quad \Big|_{\omega = k\omega_0}$$

$$= \frac{-1}{j\omega+4} e^{-(j\omega+4)\tau} \Big|_0^{+\infty} = \frac{1}{j\omega+4}$$

$$|H(\omega)| = \frac{1}{j\omega+4} \quad \Big|_{\omega = k\omega_0 = k2\pi}$$

$$k = 2, -2, 3, -3$$

$$b_2 = c_2 H(j2 \cdot 2\pi)$$